

How do current surgical pathology classification systems categorize and document benign thyroid pathologies in operative outcomes research?

Current surgical pathology classification systems, predominantly the Bethesda System for Reporting Thyroid Cytopathology, categorize benign thyroid pathologies into a standardized six-tier framework where Category II encompasses entities such as nodular goiter, follicular adenoma, and thyroiditis, though operative outcomes research consistently demonstrates that actual false-negative rates in surgical populations (3-13%) exceed the system's predicted malignancy risk of 0-3%, reflecting both selection bias in surgical series and substantial inter-observer variability in cytopathological interpretation.

Abstract

This systematic review of 40 studies reveals that the Bethesda System for Reporting Thyroid Cytopathology serves as the dominant classification framework for categorizing benign thyroid pathologies in operative outcomes research, with regional adaptations including the Italian SIAPEC classification and UK RCPATH guidelines. The benign category (Bethesda II) encompasses multiple histopathological entities including nodular/colloid goiter (39-81.7%), follicular adenoma (3-54%), and Hashimoto's thyroiditis (7.9-33.3%). The introduction of noninvasive follicular thyroid neoplasm with papillary-like nuclear features (NIFTP) has substantially reduced reported malignancy rates across Bethesda categories, with pooled risk differences of 5.5% and the greatest impact on indeterminate categories.

However, the evidence demonstrates significant limitations in how these classification systems perform in surgical populations. False-negative rates for benign cytological diagnoses in surgical series (3.0-13.4%) consistently exceed the 0-3% predicted by the Bethesda System, reflecting selection bias toward patients with concerning clinical or imaging features. Inter-observer concordance for cytopathology reaches only 64% between local and central pathologists, with central experts making fewer indeterminate diagnoses (41.2% vs 55.0%). Thyroiditis has been identified as a confounding factor that may lead to overstatement of malignancy risk. These findings indicate that while current classification systems provide standardized terminology and risk stratification, their predictive accuracy varies substantially based on institutional expertise, patient selection criteria, and the presence of confounding pathologies.

Paper search

We performed a semantic search using the query "How do current surgical pathology classification systems categorize and document benign thyroid pathologies in operative outcomes research?" across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 500 papers most relevant to the query.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Benign Thyroid Pathology Population:** Does this study involve patients with benign thyroid pathologies (excluding studies that focus exclusively on malignant thyroid conditions)?
- **Surgical Pathology Classification System:** Does this study utilize or evaluate surgical pathology classification systems (such as WHO classification, Bethesda System, or institutional classification schemes)?
- **Operative Outcomes with Pathological Documentation:** Does this study report operative outcomes or surgical results with accompanying pathological documentation?

- **Appropriate Study Design:** Is this study a systematic review, meta-analysis, randomized controlled trial, or observational study (cohort, case-control, or cross-sectional)?
- **Adequate Sample Size:** Does this study have an adequate sample size (i.e., is it NOT a case report or case series with fewer than 10 patients)?
- **Histopathological Classification Focus:** Does this study focus on histopathological classification (i.e., is it NOT exclusively focused on cytological, molecular, or genetic classification without histopathological correlation)?
- **Human Studies:** Is this a human study (i.e., NOT an animal study or in vitro research)?
- **Peer-Reviewed Publication:** Is this a peer-reviewed publication with sufficient methodological detail (i.e., NOT a conference abstract, editorial, or opinion piece)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Classification Systems:**

Extract all pathological classification systems mentioned or used in the study including:

- Name of classification system (e.g., Bethesda System, institutional systems)
- Year implemented or timeframe of use
- Whether it was used for pre-operative cytology, post-operative pathology, or both
- Any modifications or adaptations to standard systems
- Comparison between different classification systems if applicable

- **Benign Pathology Categories:**

Extract how benign thyroid pathologies are specifically categorized within the classification systems including:

- Specific category names used for benign lesions (e.g., 'benign', 'follicular adenoma', 'Bethesda II')
- Subcategories or subtypes of benign pathologies
- Diagnostic criteria or definitions provided for benign categories
- Any evolution or changes in categorization over time
- Indeterminate categories that may include benign lesions

- **Documented Benign Pathologies:**

List all specific benign thyroid pathological diagnoses documented in the study including:

- Histopathological diagnosis names (e.g., follicular adenoma, Hashimoto's thyroiditis, multinodular goiter, Graves' disease)
- Frequency or prevalence of each benign pathology type
- Incidental benign findings
- Size criteria or characteristics mentioned for benign lesions

- **Study Design:**

Extract study methodology details including:

- Study design type (retrospective, prospective, cohort, case series, etc.)
- Time period of data collection

- Single institution vs multi-center
- Sample size and patient selection criteria
- Follow-up duration if applicable

- **Operative Outcomes:**

Extract all surgical/operative outcomes measured in relation to benign pathologies including:

- Type of surgical procedures performed (partial thyroidectomy, total thyroidectomy, etc.)
- Correlation between pre-operative classification and final pathology
- False positive/negative rates for benign diagnoses
- Incidental malignancy rates in benign cases
- Long-term outcomes or complications
- Decision-making factors for surgical vs non-surgical management

- **Documentation Methods:**

Extract information about how pathological findings are documented and reported including:

- Reporting formats or templates used
- Integration with clinical decision-making
- Quality assurance or standardization measures
- Inter-observer variability or consistency measures
- Electronic vs paper documentation systems
- Correlation between cytology and histopathology reporting

- **Clinical Setting:**

Extract details about the clinical context including:

- Type of healthcare institution (academic medical center, community hospital, etc.)
- Geographic location if relevant
- Specialties involved (pathology, surgery, endocrinology)
- Patient population characteristics (demographics, referral patterns)
- Healthcare system context that might affect classification practices

Characteristics of Included Studies

This systematic review synthesizes findings from 40 studies examining pathological classification systems for benign thyroid pathologies in operative outcomes research. The studies span multiple continents and healthcare settings, with publication dates ranging from 1995 to 2025.

Study	Full Text Retrieved?	Study Design	Sample Size	Setting	Geographic Location
B. Richmond et al., 2014	Yes	Retrospective	180 patients	Academic medical center	Charleston, West Virginia
E. Cibas et al., 2013	No	Prospective	653 patients	Multi-center (academic and community)	Not specified

Study	Full Text Retrieved?	Study Design	Sample Size	Setting	Geographic Location
M. Pastoricchio et al., 2020	Yes	Retrospective	119 patients	Academic medical center	Trieste, Italy
Brenessa Lindeman et al., 2018	No	Retrospective cohort	353 patients	Tertiary-care hospital	Not specified
C. Rana et al., 2021	No	Retrospective	Not specified	Not specified	India
P. Del Rio et al., 2018	No	Retrospective	637 patients	Academic medical center	Parma, Italy
Komal Gupta et al., 2024	No	Retrospective	124 patients	Not specified	Not specified
Adnan Özpek et al., 2019	No	Retrospective	361 patients	Not specified	Not specified
M. Kihara et al., 2021	No	Retrospective	3,102 patients	Single institution	Not specified
Jiamin Chen et al., 2023	No	Retrospective	227 nodules	Single institution	Not specified
Saleh Al-wajih et al., 2020	Yes	Retrospective	260 patients	Academic medical center	Sana'a, Yemen
G. Conzo et al., 2013	No	Retrospective	472 patients	Tertiary care center	Not specified
S. Elankumar et al., 2013	No	Prospective observational	109 patients	Academic medical center	Coimbatore, India
Yunus Acar et al., 2017	No	Retrospective	542 patients	Single institution	Not specified
Dawn P. Bradley et al., 2009	No	Retrospective	678 patients	Single institution	Not specified
M. Kihara et al., 2011	No	Retrospective	445 patients	Single institution	Not specified
M. Bongiovanni et al., 2019	No	Systematic review/meta-analysis	13,752 thyroidectomies	Literature review	Not applicable
M. Goldfarb et al., 2011	No	Retrospective	624 patients	Not specified	Not specified
J. Lew et al., 2011	No	Retrospective	797 patients	Academic medical center	Not specified
L. K. Nithya et al. (n.d.)	No	Prospective case series	100 patients	Government tertiary care hospital	Visakhapatnam, India
Champa Sushel et al., 2009	No	Descriptive	140 specimens	Academic medical center	Hyderabad
Samuel J Trosman et al., 2017	No	Retrospective	74 patients	Tertiary care academic center	Not specified

Study	Full Text Retrieved?	Study Design	Sample Size	Setting	Geographic Location
Ali Koyuncuer et al., 2016	No	Retrospective	1,149 patients	Not specified	Not specified
Pietro Accetta et al., 2011	No	Retrospective	66 patients	Not specified	Not specified
Levon Katsakhyan et al., 2020	No	Retrospective	1,508 patients	Single institution	Not specified
Melissa L. Mao et al., 2018	No	Retrospective	847 patients	Single institution	Not specified
M. Bahl et al., 2014	No	Retrospective	47 patients	Single center	Not specified
S. Agarwal et al., 2019	No	Retrospective	Not specified	Multi-institutional	6 Asian countries
S. Ocak et al., 2019	No	Retrospective	366 patients	Academic medical center	Samsun, Turkey
Mário Sérgio Rocha Macedo et al., 2025	No	Retrospective	91 patients	Not specified	Not specified
A. Renshaw et al., 2017	No	Retrospective	12,764 aspirates	Single institution	Not specified
Y. Hirayama et al., 2023	No	Retrospective audit	132 patients	District General Hospital	East of England
Herb Chen et al., 1995	No	Retrospective	125 patients	Academic medical center	Baltimore, Maryland (implied)
Remmi S. Singh et al., 2011	No	Retrospective	Not specified	Academic medical center	Boston (implied)
X. Gan et al., 2021	No	Systematic review/meta-analysis	20,118 cases	Not applicable	Not specified
D. Januś et al., 2025	Yes	Retrospective	47 patients	University Children's Hospital	Krakow, Poland
Thamer Alghamdi et al., 2021	No	Retrospective	277 patients	Single center	Saudi Arabia
Ryan P. Lau et al., 2017	No	Retrospective	750 FNAs	Large academic institution	Not specified
M. Barczyński et al., 2011	No	Retrospective cohort	8,032 patients	Single institution	Not specified
K. Chaitanya et al., 2020	No	Prospective observational	78 patients	Tertiary health center	Not specified

The included studies are predominantly retrospective in design (35 of 40 studies), with only four prospective studies and two systematic reviews with meta-analyses . Sample sizes ranged from 47 patients to over 20,000 cases , with most studies conducted at single academic institutions. Geographic representation spans North America, Europe, Asia, and the Middle East, providing diverse healthcare system perspectives on classification practices.

Classification Systems for Benign Thyroid Pathologies

Primary Classification Systems Identified

Classification System	Studies Using System	Application	Key Features
Bethesda System for Reporting Thyroid Cytopathology (BSRTC)	25+ studies	Pre-operative cytology	Six-category system developed 2007
Italian SIAPEC Classification	1 study	Pre-operative cytology	TIR3A/TIR3B subcategories for indeterminate lesions
ACR TI-RADS	1 study	Pre-operative imaging	Risk stratification for imaging findings
RCPATH (UK)	2 studies	Pre-operative cytology	THY categories including THY3a/THY3f
WHO Classification of Thyroid Tumors	1 study	Post-operative pathology	Updated 2022 with borderline tumor categories
Institutional Systems	Multiple studies	Variable	Hospital-specific ultrasonographic classifications

The Bethesda System for Reporting Thyroid Cytopathology represents the dominant classification framework across the reviewed literature, implemented in 2007 and subsequently adopted internationally. The system divides cytological findings into six categories: non-diagnostic (I), benign (II), atypia of undetermined significance/follicular lesion of undetermined significance (III), follicular neoplasm/suspicious for follicular neoplasm (IV), suspicious for malignancy (V), and malignant (VI) . One prospective multicenter study demonstrated that using the six-category Bethesda System, 64.0% of diagnoses made by local and central cytopathologists were concordant, with 74.7% intraobserver concordance .

Regional variations exist in classification approaches. The Italian SIAPEC classification, modified in 2014, introduced TIR3A (low-risk indeterminate) and TIR3B (high-risk indeterminate) subcategories, with TIR3B including cases with mild or focal nuclear atypia suggestive of papillary carcinoma . This system demonstrated improved diagnostic accuracy and clinical management compared to standard Bethesda categories . Similarly, the Royal College of Pathologists guidelines divide indeterminate THY3 categories into THY3a (atypia/non-diagnostic) and THY3f (follicular lesion/suspect follicular neoplasia) .

Categorization of Benign Lesions Within Classification Systems

Bethesda Category	Description	Expected ROM	Studies Reporting
Category II (Benign)	Benign cytological findings	0-3%	Multiple
Category III (AUS/FLUS)	Atypia of undetermined significance	Variable	Multiple
Category IV (FN/SFN)	Follicular neoplasm/suspicious for follicular neoplasm	15-30%	Multiple

The benign category (Bethesda II) encompasses multiple histopathological entities including adenomatoid or hyperplastic nodules , multinodular goiter , Hashimoto's thyroiditis , colloid nodules , and follicular adenoma . Indeterminate categories (III and IV) may contain lesions that prove benign on final pathology, representing a significant clinical challenge .

Impact of NIFTP Reclassification

The introduction of noninvasive follicular thyroid neoplasm with papillary-like nuclear features (NIFTP) as a distinct entity with low malignant potential has substantially impacted classification practices . Multiple studies documented the effects of this nomenclature change:

Study	NIFTP Prevalence	ROM Change (Overall)	Most Affected Categories
Lindeman et al., 2018	7.3%	-7.3%	Bethesda III (-15%), V (-18%)
Rana et al., 2021	6.9% of excised nodules	Variable by category	Indeterminate categories
Mao et al., 2018	22% of FVPTC cases	-3% overall	Bethesda III-V
Lau et al., 2017	11.6% of resections	Variable	AUS/FLUS (-20%), FN/SFN (-30.8%)
Katsakhyan et al., 2020	9.5% of thyroidectomies	Malignant diagnoses decreased from 54.5% to 44.6%	Variable
Bongiovanni et al., 2019	Meta-analysis pooled data	-5.5% pooled risk difference	Bethesda V (-14%), VI (-3%)
Renshaw et al., 2017	Not specified	Benign malignancy rate: 4.3% to 2.4%	Benign category

The NIFTP reclassification has significantly reduced the risk of malignancy across Bethesda categories, particularly affecting indeterminate diagnoses. One study from India reported that inability to diagnose NIFTP preoperatively led to overtreatment in 16.2% of cases , emphasizing the clinical importance of this classification change. The prevalence of NIFTP in Asian countries may be higher than initially expected .

Documented Benign Thyroid Pathologies

Histopathological Diagnoses and Prevalence

Benign Pathology	Prevalence Range	Studies Reporting
Nodular/colloid goiter	39-81.7%	Multiple
Follicular adenoma	3-54%	Multiple
Hashimoto's thyroiditis	7.9-33.3%	Multiple
Adenomatous hyperplasia	13.7-45%	Multiple
Multinodular goiter	39-56.1%	Multiple
Autoimmune thyroiditis	33.3%	Limited
Hurthle cell adenoma	3%	Limited
Fibrosis	45.4%	Limited

Nodular hyperplasia and colloid goiter consistently represent the most common benign pathologies across studies, accounting for up to 81.7% of cases in some series . One study reported adenomatous goiter as the most common benign lesion, comprising 60% of all thyroid nodules . Follicular adenoma prevalence varies substantially between studies, ranging from 3% to 54% , likely reflecting differences in patient selection criteria and referral patterns.

Histopathological features correlating with suspicious ultrasound appearances were examined in one study of 227 benign nodules classified as ACR TI-RADS 4 or 5 . Fibrosis was the most common histopathological feature in both categories (40.8% in TI-RADS 4, 49.2% in TI-RADS 5) . Thyroid nodules with fibrosis demonstrated sonographic features of "taller than wide," while lesions with follicular epithelial hyperplasia were more likely to present with irregular margins and very hypoechoic appearances .

Benign Pathologies in Pediatric Populations

One pediatric-focused study documented specific benign entities including thyroid follicular nodular disease (32.1%), thyroid follicular adenoma (9.5%), oncocytic adenoma (1.5%), dyshormonogenetic goiter (0.8%), and therapy-resistant Graves' disease (8.0%) . The WHO 2022 Classification introduced the term "follicular nodular disease" to better characterize certain benign lesions , and borderline tumor categories (FT-UMP, WDT-UMP) were established to address lesions with uncertain malignant potential .

Operative Outcomes Related to Benign Pathology Classification

False Negative Rates and Incidental Malignancy

Study	Population	False Negative Rate	Incidental Malignancy Rate
Richmond et al., 2014	Benign FNAB patients	13.4%	6.7% microcarcinomas, 6.7% significant cancers
Lew et al., 2011	Dominant nodules >1cm	8.6%	8.6%
Kihara et al., 2021	Cytologically benign tumors	3.0%	3.0%
Özpek et al., 2019	Benign MNG patients	Not specified	11.6%
Kihara et al., 2011	Benign/follicular nodules	11.6% (immediate), 6.4% (delayed)	Same as false negative
Bradly et al., 2009	Benign thyroid diseases	Not specified	12% overall IPC

Study	Population	False Negative Rate	Incidental Malignancy Rate
Elankumar et al., 2013	Benign thyroid conditions	Not specified	6.4%
Koyuncuer et al., 2016	Nodular goiter patients	Not specified	5.39%
Barczyński et al., 2011	BBTD patients	Not specified	5.0%
Nithya et al. (n.d.)	Benign thyroid mass lesions	Not specified	9.09%
Hirayama et al., 2023	U1/U2 nodules	Not specified	15%

The false negative rate for benign cytological diagnoses ranged from 3.0% to 13.4% , substantially higher than the 0-3% predicted by the Bethesda System . This discrepancy between expected and observed malignancy rates in surgical series has important implications for patient counseling and management decisions.

Incidental papillary carcinoma rates in patients operated for presumed benign disease ranged from 5.0% to 15% . One study specifically examining incidental thyroid carcinoma in multinodular goiter patients found papillary thyroid microcarcinoma in 10.5% and papillary thyroid carcinoma >1cm in 1.1% of cases . Hashimoto's thyroiditis demonstrated the highest rate of incidental papillary carcinoma among benign thyroid diseases, with contralateral involvement observed in 40% of patients with follicular adenoma .

Surgical Procedures and Decision-Making

Procedure Type	Indications	Studies Documenting
Hemithyroidectomy	Solitary low-risk lesions , younger patients	Multiple
Total thyroidectomy	Bilateral disease , high-risk features	Multiple
Completion thyroidectomy	Malignancy on final pathology	Multiple

The choice between hemithyroidectomy and total thyroidectomy for benign disease remains controversial. One large retrospective study of 8,032 patients compared bilateral subtotal thyroidectomy with total thyroidectomy and found that subtotal thyroidectomy resulted in significantly higher rates of completion thyroidectomy for incidentally diagnosed thyroid cancer (2.15% vs 0.10%, $p<0.001$) and reoperation for recurrent goiter (45.33% of 6.99% with recurrence vs 0% after total thyroidectomy, $p<0.001$) . However, total thyroidectomy was associated with higher rates of transient hypoparathyroidism (13.12% vs 2.70%, $p<0.001$), though permanent hypoparathyroidism rates were similar (0.10% vs 0.15%, $p=0.65$) .

Hemithyroidectomy was performed more frequently in younger patients and in those with NIFTP diagnoses compared to invasive follicular variant papillary thyroid carcinoma (77.14% vs 51.69%, $p=0.01$) . Patients with Bethesda category V or VI cytology were more likely to undergo total thyroidectomy .

Correlation Between Classification and Operative Outcomes

Bethesda Category	Reported Malignancy Rates	Studies
I (Non-diagnostic)	13.3-37.5%	Multiple
II (Benign)	2.8-9.6%	Multiple
III (AUS/FLUS)	7.3-42.3%	Multiple
IV (FN/SFN)	15.5-48.7%	Multiple
V (SFM)	61.7-93.6%	Multiple
VI (Malignant)	96.5-100%	Multiple

Reported malignancy rates varied considerably across institutions and study populations. The wide range within each Bethesda category reflects differences in patient selection, surgical indications, and local pathology practices. Studies conducted before adoption of the Bethesda system showed similar overall malignancy rates (46%) but used different categorization schemes .

One study specifically examining the correlation between BSRTC categories and actual outcomes found higher malignancy rates in Bethesda II, III, and IV categories than predicted by the system, particularly in patients with concurrent thyroiditis . Thyroiditis was identified as a confounding factor in cytological examination that may lead to overstatement of malignancy risk .

Documentation and Quality Assurance Practices

Standardization Through Classification Systems

The Bethesda System serves as the primary framework for standardized documentation , providing a common language among pathologists and surgeons . Implementation of the Bethesda classification led to significant reductions in lesions reported as "benign" (from 45% to 32.9%) and "non-diagnostic" (from 16.4% to 4.7%) without compromising actual malignancy detection rates .

Inter-observer variability remains a recognized limitation of visual microscopic diagnosis. One prospective multicenter study found that concordance on the histopathologic distinction between benign and malignant diagnoses was 91% between local and central histopathologists and 90% between two central histopathologists . For cytopathology using the Bethesda System, only 64.0% of diagnoses were concordant between local and central cytopathologists . Central cytopathologists made fewer indeterminate diagnoses than local pathologists (41.2% vs 55.0%) .

Integration with Clinical Decision-Making

Multiple studies documented the integration of pathological classification with clinical decision-making pathways:

- Ultrasonographic findings were identified as important for therapeutic decision-making, with significant associations between suspicious ultrasound appearance and underlying carcinoma ($p=0.004$)
- Molecular markers (BRAF, RAS mutations) complement morphologic classification, with BRAF mutation showing 89% predictability for malignancy compared to 47% for RAS mutation
- FNAC and USG findings guide surgical intervention decisions, with individualized patient management recommended
- Compliance with repeat FNAC requirements (84%) and multidisciplinary team discussions (97%) were documented in one audit

Cytology-Histopathology Correlation

Study	Concordance Rate	Notable Findings
Al-wajih et al., 2020	88.07%	Lower accuracy for follicular carcinoma (40%)
Koyuncuer et al., 2016 Cibas et al., 2013	55.7% concordant, 3% discordant 91% for benign/malignant distinction	85 cases excluded as non-diagnostic Substantial variability for intermediate categories

The correlation between preoperative cytology and final histopathology varies by lesion type. FNAB demonstrated sensitivity ranging from 70.8% to 88.8% and specificity from 97.1% to 99.9% , though accuracy was notably lower for follicular carcinoma at 40% .

Synthesis

The heterogeneity in reported malignancy rates across classification categories reflects several systematic factors rather than random variation. Studies with higher false-negative rates for benign diagnoses (>10%) were predominantly surgical series with specific selection criteria—patients proceeded to surgery despite benign cytology due to symptoms, nodule enlargement, or concerning imaging features . In contrast, studies reporting lower false-negative rates (~3%) followed broader populations including those observed without surgery .

The discrepancy between Bethesda-predicted malignancy rates and observed rates in surgical series can be reconciled by recognizing selection bias: surgical series inherently exclude patients whose benign cytology was accepted and who did not undergo surgery. The true false-negative rate in unselected populations remains uncertain, but the surgical literature consistently demonstrates that patients selected for operation despite benign cytology face higher malignancy risk (6-15%) than the 0-3% suggested for all benign diagnoses .

Geographic variation in NIFTP prevalence (6.9-22% of applicable cases) may reflect genuine population differences or variations in pathological interpretation criteria. Asian populations may have higher NIFTP prevalence than initially expected based on Western data , though institutional practices in applying NIFTP diagnostic criteria also contribute to observed variation.

The impact of thyroiditis on classification accuracy represents a distinct confounder. Studies specifically examining patients with concurrent thyroiditis found higher than expected malignancy rates in lower Bethesda categories , while Hashimoto's thyroiditis was associated with the highest incidental papillary carcinoma rates among benign diseases . The inflammatory changes associated with thyroiditis may produce cytological atypia that mimics malignancy, leading to both false-positive indeterminate diagnoses and potentially obscured true malignancies.

Institutional experience and pathologist expertise significantly influence classification outcomes. Central expert review in one multicenter study reduced indeterminate diagnoses from 55.0% to 41.2% , suggesting that increased experience leads to more definitive categorization. This finding has implications for centers with lower thyroid cytology volumes, where indeterminate diagnoses may be proportionally higher.

For clinical decision-making regarding patients with benign cytology who have concerning features, the evidence supports that suspicious ultrasound appearance , larger nodule size , and presence of thyroiditis warrant consideration of surgical intervention despite benign FNAC results. Patients without these risk factors and with reassuring

ultrasound characteristics (isoechoic, cystic component, regular borders, no microcalcifications) have a very low malignancy risk (1.5% false-negative rate) and may be suitable for observation .

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